CTA200 Assignment 3

Wendy Tso

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1 Introduction

This assignment studies two nonlinear dynamical systems: the Mandelbrot set and the Lorenz system. Both systems demonstrate how simple deterministic rules can generate complex and chaotic behavior.

2 Question 1: Mandelbrot Set

The Mandelbrot set is defined by iterating the complex recurrence relation

$$z_{n+1} = z_n^2 + c,$$

where $c = x + iy$ is a complex parameter and $z_0 = 0$.

For each point c in the complex plane, we determine whether the sequence remains bounded or diverges. Points that remain bounded are classified as part of the Mandelbrot set.

Two visualizations are produced: (i) a binary plot showing bounded vs divergent points, and (ii) an escape-time plot showing the iteration at which divergence occurs.

3 Question 2: Lorenz System

The Lorenz system is given by:

$$\dot{X} = -\sigma(X - Y), \quad \dot{Y} = rX - Y - XZ, \quad \dot{Z} = -bZ + XY.$$

We numerically solve this system using `solve_ivp` with parameters $\sigma = 10$, $r = 28$, and $b = 8/3$. The system is known to exhibit chaotic dynamics and strong sensitivity to initial conditions.

Results: Question 1

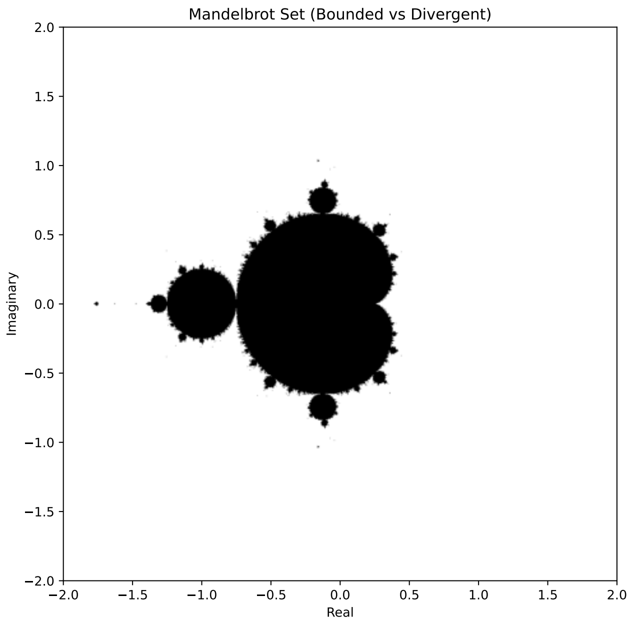

Figure 1: Mandelbrot set: bounded vs divergent points.

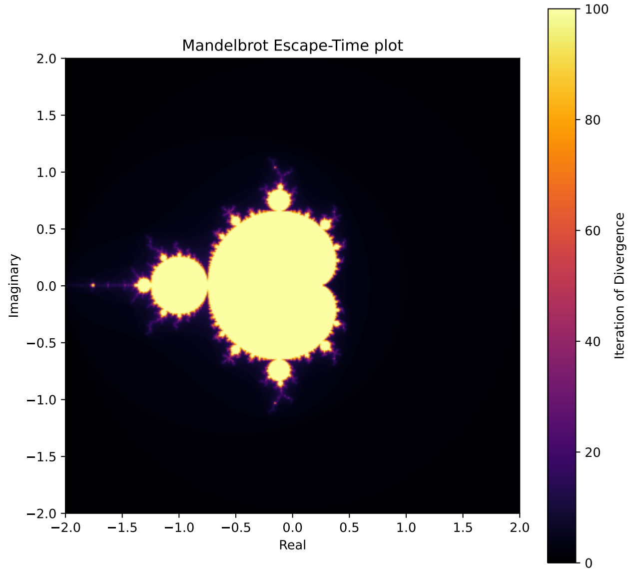

Figure 2: Mandelbrot escape-time plot.

Results: Question 2

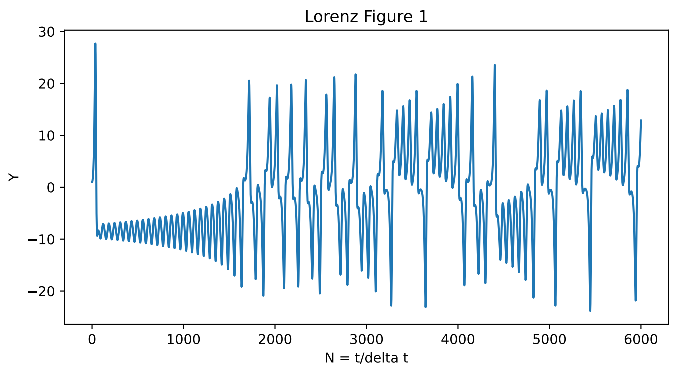

Figure 3: Time evolution of the Lorenz system (variable Y).

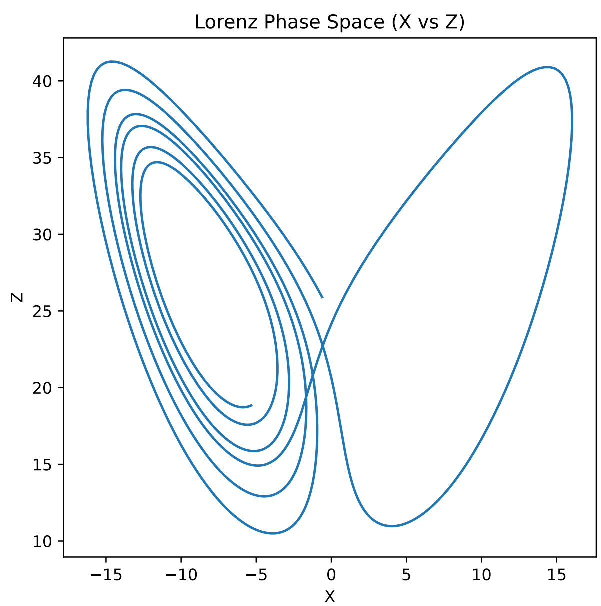

Figure 4: Lorenz phase space projection (X vs Z).

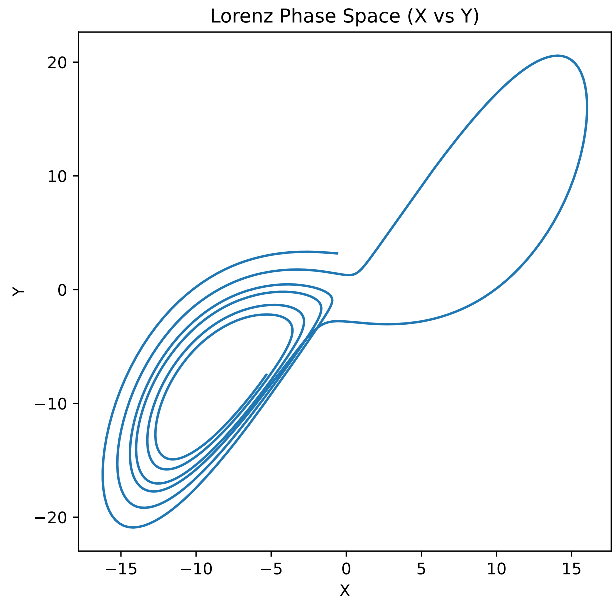

Figure 5: Lorenz phase space projection (X vs Y).

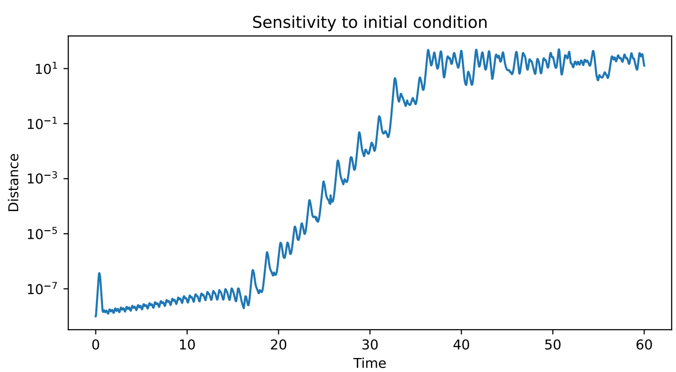

Figure 6: Sensitivity to initial conditions showing exponential divergence.